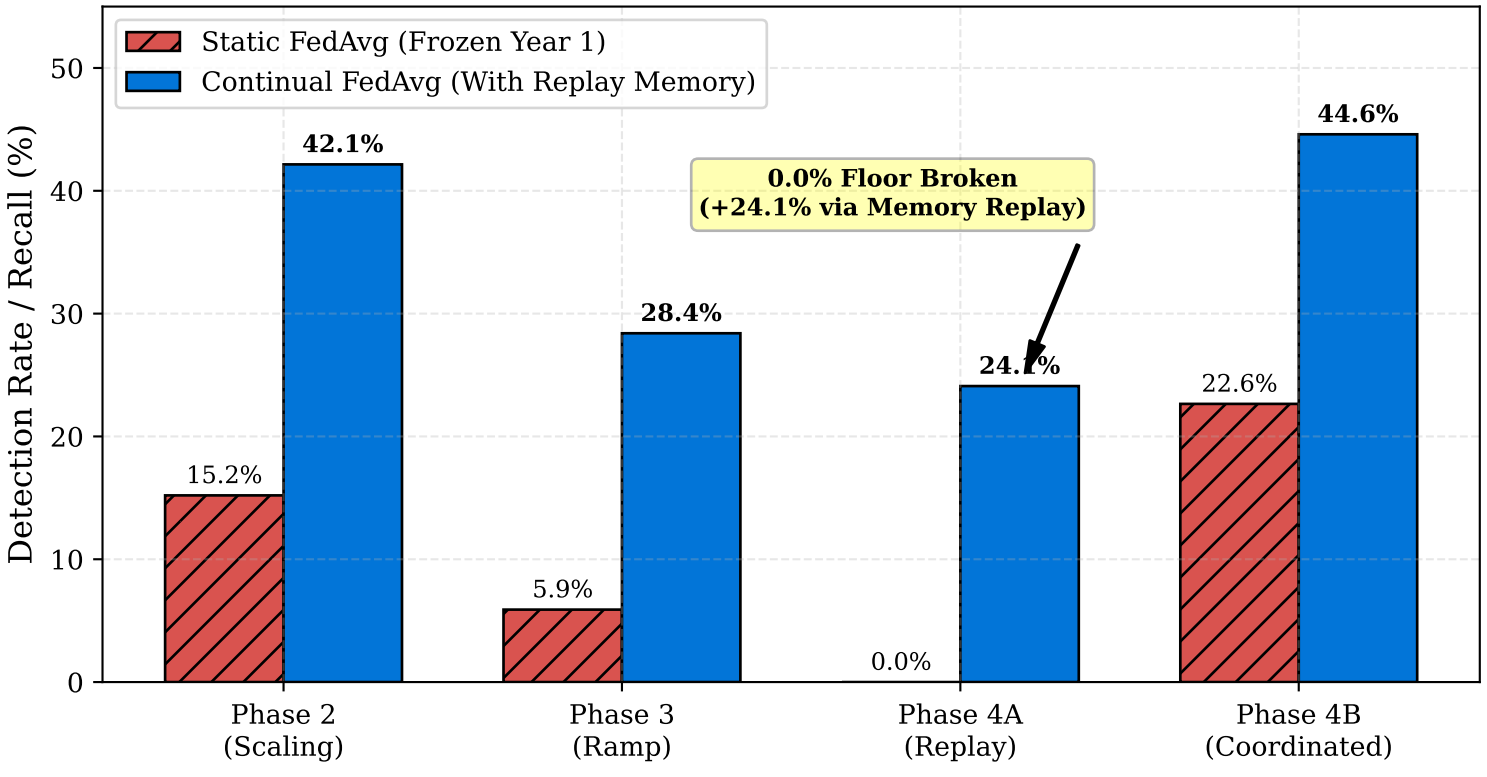
Protocol A: Detection Across Evolving Attack Phases (All 299 Prosumers)
Static FedAvg (Frozen Year 1)
Continual FedAvg (With Replay Memory)
Detection Rate / Recall (%)
0
10
20
30
40
50
42.1%
15.2%
28.4%
5.9%
0.0%
24.1%
22.6%
44.6%
0.0% Floor Broken
(+24.1% via Memory Replay)
Phase 2
(Scaling)
Phase 3
(Ramp)
Phase 4A
(Replay)
Phase 4B
(Coordinated)